

EM616: Time-Series Analysis

Introduction

Arun K. Tangirala

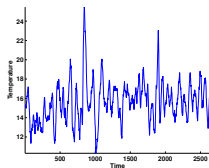
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Time-Series

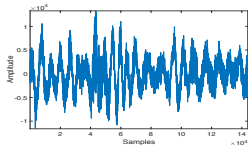
A time-series simply refers to an **ordered** collection of data (usually in time)

- ▶ e.g., yearly wages, annual production, daily temperature, hourly satellite images
- ▶ Measurements could be a function of other dimensions (e.g., frequency, space)
- ▶ Data may be collected at regular or irregular intervals
- ▶ Many variables could be recorded simultaneously (multivariate data)

Examples



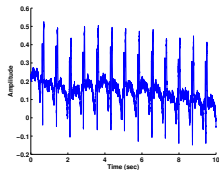
(a) Temperature



(b) Seismic noise

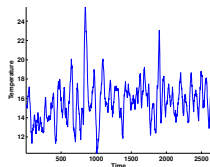


(c) Swiss SMI

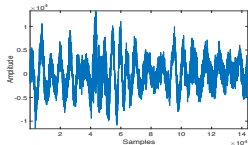


(d) ECG

Examples



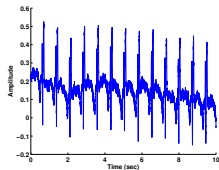
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Time-series can consist of different components:

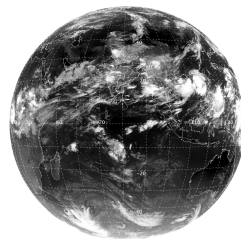
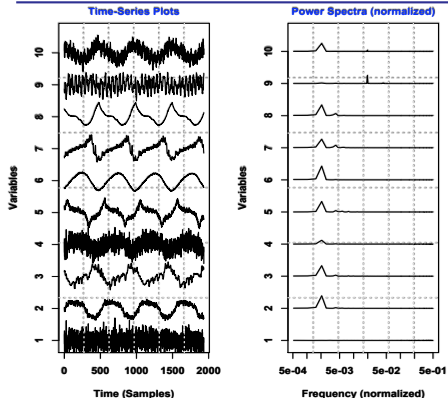
► Deterministic

- 1 Trend (polynomial, spline)
- 2 Seasonal (periodic) and Cyclic components

► Stochastic

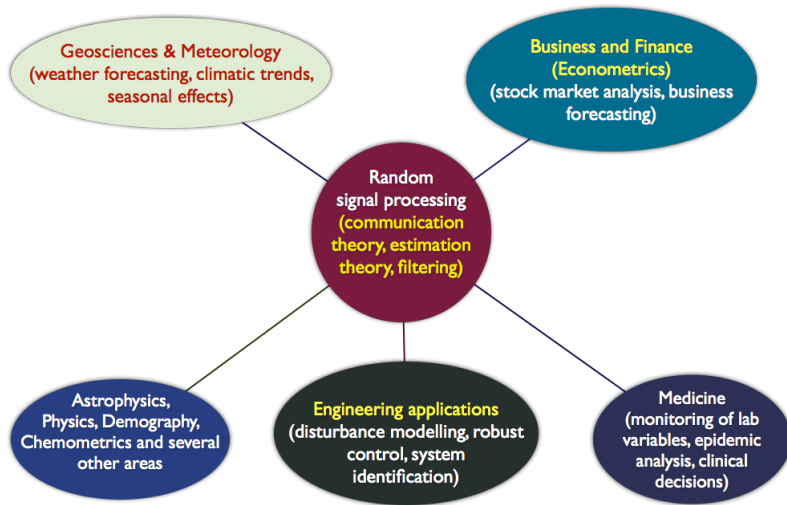
- 1 Stationary component
- 2 Non-stationary trend (e.g., random walk)
- 3 Changing mean, variance, temporal correlation, etc.

More examples

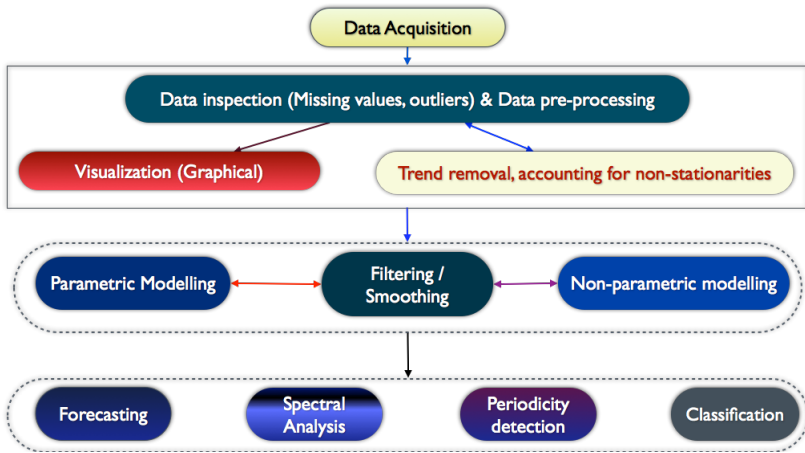


Time-series could be **multivariate**, **multidimensional** and also **transformed** into another domain

Applications



Systematic analysis of time-series data



Challenges in time-series analysis

Uncertainty (Randomness)

- i. Lack of an exact mathematical function to describe the process of interest (leading to a **probabilistic** framework).
- ii. To be able to draw inferences about the ensemble from a single realization.
- iii. Estimation of “unknowns” from (uncertain) observations / knowns.

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Sources of uncertainty:

- 1 Process understanding (modelling errors)
- 2 Unknown causes (factors influencing the variable and disturbances)
- 3 Measurement errors

The “realization” challenge

Central premise

A **time-series** is an **ordered collection of random variables**, i.e., at each instant the observation is a random variable. For multivariate time-series, it is a collection of random variables.

A single time-series is said to be a **realization** of the random process.

The “realization” challenge

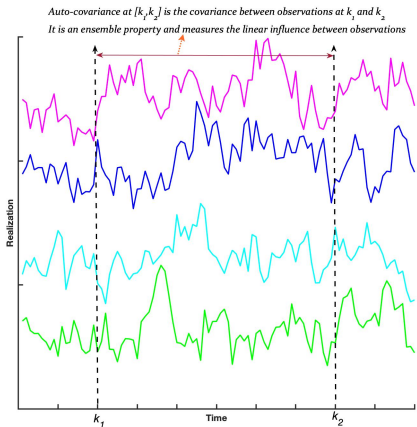
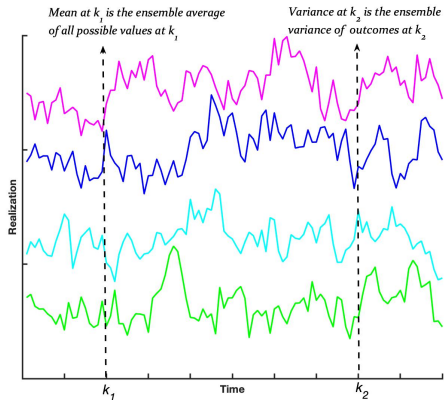
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The collection of all possible realizations is said to be the **ensemble**.

Random Process: Ensemble of realizations



Properties of any random signal (process) are its **ensemble** properties and can change with time.

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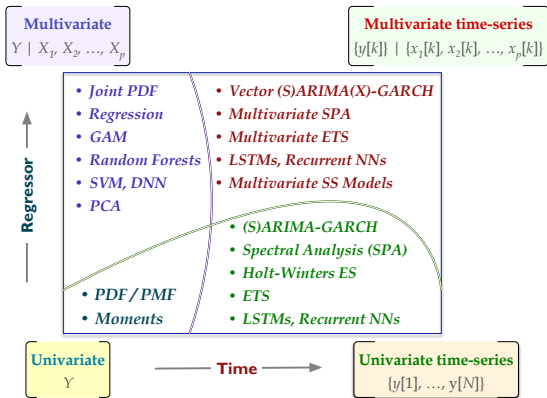
Two questions that repeatedly arise:

- 1 **How good are the estimates** (of model parameters, statistical properties, etc.) from a single realization?
- 2 **Is a single estimate sufficient?**

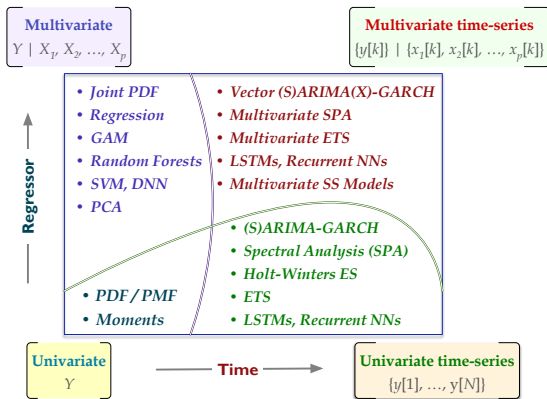
Example: The average reading of a variable from sensor 1 is 3.005 and from sensor 3, is 2.996. What can we say about the true level?

Four important streets and corners

- 1 How do we predict an RV from its distribution (probability to prediction)? (rarely discussed)

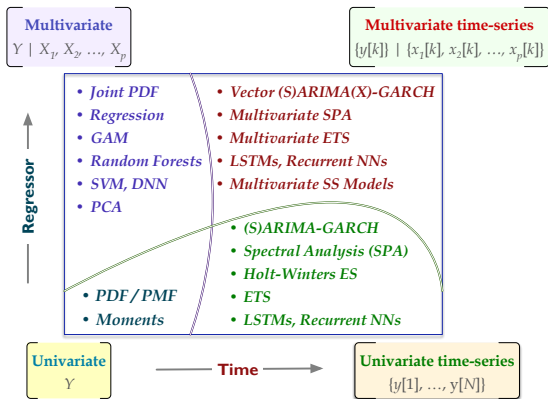


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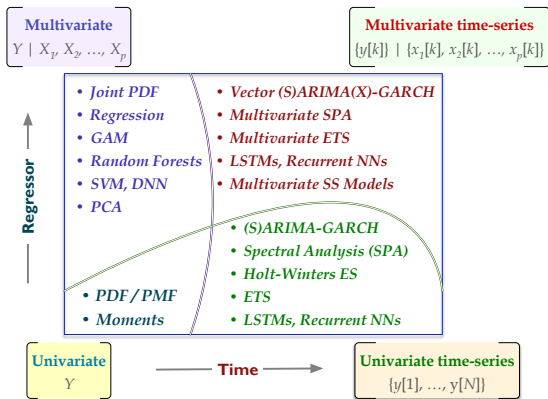
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- 2 Given the knowledge of explanatory variables, what is the best prediction? (multivariate DA)
- 3 Suppose the history (time-series) of RV is given. Does the prediction improve? If yes, how do we predict? (univariate TSA)
- 4 Finally, if both the explanatory variables and their time history are provided, what is the best prediction? (multivariate TSA)

Estimation

Estimators are at the heart of TSA. They produce estimates of “unobserved” or “hidden” quantities / variables from observations.

We shall learn how to:

- ▶ Characterize “goodness” of estimators
- ▶ Estimate statistical properties / parameters / signals
- ▶ Report estimation results and test hypotheses (on random processes)

A few remarks

- ▶ Emphasis is on **concepts** rather than on the rigour.
- ▶ Idea is to introduce and illustrate ideas first through examples.
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- ▶ TSA requires **both theory and skill**, i.e., it is both a science and an art.
- ▶ A computational tool is indispensable to a good understanding and practice.

R: Software for TSA

R[®] is an integrated software package for data manipulation, calculation and graphical display

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- ▶ Writing user-defined and specialized add-on packages is easy
- ▶ Use of **RStudio**[®] makes the use of R easy and useR-friendly!

Useful references / pre-requisites

- ➊ **Introduction to Statistical Hypothesis Testing** as contained in the video lectures of **NPTEL / MOOC** (<http://www.nptel.ac.in/courses/103106120/>)
- ➋ Basics of linear and matrix algebra.
- ➌ **Probability Distributions (Densities) and Moments** as contained in **Lectures 01A - 09B** of **NPTEL / MOOC** video lectures on **ATSA** (<http://www.nptel.ac.in/courses/103106123/>).
- ➍ **R Tutorials for Beginners**
 - ▶ <https://youtu.be/1mQIEdhPv6I>
 - ▶ <https://youtu.be/hvXr2MHIexQ>

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